

Identifying Patients at Risk of Sudden Cardiac Arrest using Graph Representation Learning on Electronic Health Records

Fatemeh Ghaffarpour
Dept. of Electrical & Software Eng.
University of Calgary
Calgary, Canada
fatemeh.ghaffarpour@ucalgary.ca

Emad A. Mohammed
Dept. of Physics & Computer Science
Wilfrid Laurier University
Waterloo, Canada
emohammed@wlu.ca

Behrouz Far
Dept. of Electrical & Software Eng.
University of Calgary
Calgary, Canada
far@ucalgary.ca

Abstract—Sudden cardiac arrest (SCA) occurs when the heart stops pumping blood due to an abnormal rhythm, making early identification of at-risk patients in the intensive care unit essential. Electronic health records (EHRs) provide multivariate time-series data, but SCA prediction remains challenging because SCA and non-SCA patients often show similar temporal patterns. Existing temporal models, such as standard long short-term memory (LSTM) networks, learn from each patient independently and may miss useful inter-patient similarity. Graph-based approaches often rely on full graph neural networks at inference, which increases computational cost. Additionally, many EHR graphs are built from static diagnosis codes rather than using time-series patient data. To address these limitations, a graph-regularized LSTM framework for SCA prediction is proposed using the MIMIC-III dataset. A bidirectional gated recurrent unit autoencoder first learns visit-level time-series representations, which are then reduced in dimension, clustered with K-means, and used to construct a cosine-similarity k -nearest neighbor graph. A graph-based regularization term is then added to the training loss, along with binary cross-entropy, to reduce distances between similar visits. Since the graph is only used during training, the final model remains a standard LSTM at test time without additional inference cost. The proposed approach achieved a PR-AUC of 0.5968 and an accuracy of 0.7937, improving over the standalone LSTM baseline.

Index Terms—Electronic Health Records, Graph Regularization, Long Short-Term Memory, Patient Similarity Graph, Sudden Cardiac Arrest

I. INTRODUCTION

Sudden cardiac arrest (SCA) is an abrupt cardiovascular collapse with rapid loss of vital signs and is a leading cause of death [1]. In Canada, SCA accounts for an estimated 35,000–45,000 deaths per year [2]. Various machine learning (ML) methods have been developed to predict SCA risk from clinical datasets [3]. Electronic health records (EHRs) provide essential information for physicians and researchers, including medication data, laboratory results, and vital signs [4]. In intensive care units (ICUs), EHRs are especially valuable because they contain rich multivariate time-series data from repeated vital signs and laboratory measurements. However, predicting SCA from EHR time-series is challenging as measurements are often collected at irregular intervals, missing data are common, and patients who experience SCA may appear physiologically similar to those who do not. These are well-known challenges in temporal EHR modeling [5]. Long short-term memory (LSTM) networks are well-suited for multivariate clinical time-series because they capture changes

over time. However, a standard LSTM learns from each visit in isolation and does not explicitly use information from similar patients. Graph-based methods can capture patient similarity, but many existing approaches rely on static data, such as diagnosis codes, rather than longitudinal data. Other graph neural network (GNN) methods require graph processing during inference, which increases computational cost. These limitations highlight the need for a method that captures patient similarity while keeping the LSTM’s ability to model temporal patterns without increasing inference cost.

This study proposes a graph-regularized deep learning (DL) framework for SCA prediction from multivariate EHR time-series data, motivated by neural graph learning (NGL) [6]. First, the method clusters patient visits according to their temporal EHR data and then constructs a similarity graph that connects similar visits within and across clusters. Within this framework, the LSTM learns temporal patterns directly from each patient’s raw multivariate time-series, whereas the graph is used only during training to guide learning through neighbor sampling and a regularization term. In this way, the method preserves the order of the original time-series and encourages visits with similar patterns to learn similar representations and predictions. The proposed approach combines within-visit temporal learning with inter-patient similarity while keeping the prediction model as a standard LSTM at inference time.

This paper is divided into five sections. Section II reviews related work on predicting SCA using EHR data. Section III discusses the dataset, research question, hypothesis, and the proposed model architecture. Section IV presents experimental results and comparisons. Finally, Section V summarizes the model’s performance, limitations, and future work.

II. LITERATURE REVIEW

Many studies have predicted cardiac arrest (CA) and heart failure (HF) from EHR data using ML, DL, and graph models.

A. Traditional ML and DL Approaches using EHR Data

A common approach is to apply classical ML classifiers to tabular features extracted from EHR time-series. Javan et al. [7] proposed ML models to predict CA in sepsis patients several hours before the event using the Medical Information Mart for Intensive Care III (MIMIC-III) database. They tested classical methods (support vector machines, decision tree, logistic regression, k -nearest neighbors, and Gaussian naive

Bayes) and ensemble methods (gradient boosting, extreme gradient boosting (XGBoost), random forest (RF), balanced bagging classifier, and stacking). The stacking model on multivariate data performed best, but the reported F1-score was relatively low. The key limitation is that converting time-series data into static feature vectors may discard temporal dynamics. Similarly, [8] compared six algorithms to predict in-hospital CA using vital signs from 1 to 12 hours before the event in MIMIC-III, and found RF performed best. Still, the temporal dimension was collapsed into a flat feature vector, losing sequential patterns. In [9], the authors used XGBoost on bedside vital signs from MIMIC-III to develop a real-time, interpretable model for predicting CA. Their model showed low area under the precision–recall curve and F1-score due to class imbalance, and it tended to misclassify non-SCA (NSCA) patients as SCA, highlighting the difficulty of distinguishing SCA from NSCA using static features. DL models were also used to model temporal sequences. In [10], the authors used an LSTM with attention mechanisms to predict 7-day mortality in cardiovascular patients based on the first 10 hours of ICU data from MIMIC-III. However, relying on receiver operating characteristic area under the curve (ROC-AUC) alone is not ideal for imbalanced prediction tasks. In [11], the authors predicted in-hospital mortality for HF patients using a deep fusion model that combined time-series and categorical MIMIC-III features. Their approach included missing-value imputation with a variational recurrent network, followed by gated recurrent unit (GRU)-based modeling and feature fusion.

B. Graph-based Approaches for EHR Modeling

Graph-based methods attempt to model relationships among patients or clinical concepts. Boll et al. [12] evaluated GraphSAGE, Graph Attention Network, and Graph Transformer on MIMIC-III EHR graphs to predict whether a patient will develop HF at the next visit. The graph was constructed using International Classification of Diseases (ICD-9) codes for diagnoses and procedures, and National Drug Codes for medications. However, because the graph contained all patients, including those in the test set, the evaluation was transductive, which complicates generalization to new patients. In [13], the authors implemented a GNN-LSTM model to predict HF outcome using a dataset of 299 patients, where the GNN learned representations from a patient-similarity graph and the LSTM modeled temporal patterns. The integrated architecture adds computational complexity, and a small dataset may increase the risk of overfitting. In addition, the use of true longitudinal time-series data was not clearly described despite the LSTM-based framework. Lyu et al. [14] introduced PDSNet, a transformer-based model with dual spatial similarity strategies for HF prediction. They constructed a disease graph based on the Clinical Modification of ICD-9 (ICD-9-CM) hierarchy and a patient-disease bipartite graph, then used a graph convolutional network and transformers to model relationships across visits. One limitation is that the model assumes diseases within the same ICD-9-CM category are clinically related. However, diseases sharing a parent code

may differ substantially in their relevance to HF. Moreover, the PDSNet model requires similarity layers during inference, which increases prediction complexity.

C. Summary and Research Gap

Traditional ML methods simplify clinical time-series into static feature vectors, often losing temporal patterns and sequential dynamics. Temporal DL models such as LSTM and GRU capture within-patient dynamics but treat each patient as an isolated sequence, without using structural similarity among patients with similar clinical trajectories. This is a bottleneck in SCA prediction, where patients with different outcomes may appear clinically similar. Graph-based models rely on static code- or feature-based similarity, and require full graph structures at inference, which increases deployment complexity and limits scalability. To address these limitations, this study proposes a graph-regularized LSTM framework for SCA prediction with four main stages. First, a bidirectional GRU autoencoder (Bi-GRU AE) [15] learns low-dimensional visit-level embeddings that represent clinical trajectories over time. Second, dimensionality reduction is applied to these embeddings, followed by K-means clustering [16] to identify latent patient phenotypes. Third, a cosine-similarity k -nearest neighbor graph is constructed to connect visits with similar temporal patterns. Fourth, graph regularization is applied during training to minimize differences between the embeddings of connected visits, encouraging the model to learn representations that reflect patient histories and shared cohort structures. Importantly, this regularization is used only during training, and the model remains a standard LSTM without additional computational cost. Additionally, pseudo-labels are generated through weighted majority voting on the patient-visit similarity graph.

III. MATERIALS AND METHODS

A. Dataset

This work uses the open-source MIMIC-III database [17], which contains records from 53,423 ICU stays at Beth Israel Deaconess Medical Center in Boston between 2001 and 2012. Since working with the raw database is challenging because each patient has multiple records, the public MIMIC-III benchmark dataset [18], a structured version of the database, is used instead. This dataset contains 17 categorical and numerical clinical variables — capillary refill rate, diastolic blood pressure, fraction inspired oxygen, Glasgow coma scale eye-opening, Glasgow coma scale motor response, Glasgow coma scale total, Glasgow coma scale verbal response, glucose, heart rate, height, mean blood pressure, oxygen saturation, respiratory rate, systolic blood pressure, temperature, weight, and pH — for 33,678 patients with 41,902 ICU stay records. These patients have multiple hospital admissions and at least one ICU stay, referred to as a visit in the benchmark. Diseases and procedures are represented using ICD-9 codes, and each visit may have multiple ICD-9 codes. Each ICU visit is represented as a multivariate time-series $\{x_t\}_{t=1}^T$, where $x_t \in \mathbb{R}^{17}$ is the vector of 17 clinical variables at time step t . To identify visits

associated with CA, three ICD-9 codes — 42741, 42742, and 4275 — are extracted from the diagnosis codes. Of the 33,678 patients, 1,265 have at least one visit with a CA code, resulting in 1,278 SCA visits and 40,624 NSCA visits.

a) Dataset Preprocessing: Preprocessing follows the MIMIC-III benchmark [18] used for the LSTM-based models. Each visit time-series is resampled into one-hour intervals using the last measurement in each interval. If no value is available, the most recent previous measurement is used; if that is missing, the normal value for that clinical variable is substituted. For each clinical variable, a mask feature is added at each time step to indicate whether the value is observed or imputed. Categorical variables are encoded using one-hot encoding. As a result, the 17 original features expand to 76 features, so each visit is represented as $\{x_t\}_{t=1}^T$, where $x_t \in \mathbb{R}^{76}$. Outlier values are handled according to [18].

b) Data Splitting Strategy: To ensure a fair split, the dataset is divided by patient ID rather than by visit. Thus, all visits from a patient belong to the training, validation, or test set. Patients with at least one SCA diagnosis are flagged as positive, and those without SCA are flagged as negative. As noted in Section III-A, 1,265 patients are labeled SCA and 32,413 are labeled NSCA. Because the dataset is imbalanced, a similar undersampling method used in [19] is applied to the patient-level split. Fig. 1 illustrates the splitting strategy. First, the 1,265 SCA patients are divided into 80% training

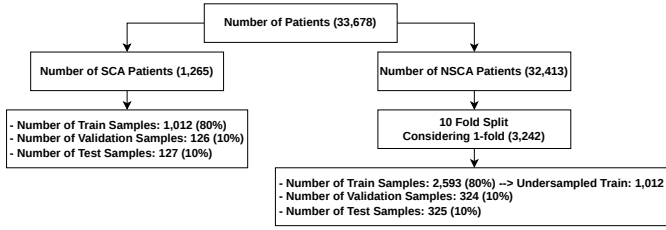


Fig. 1. Dataset splitting strategy for the training, validation, and test sets.

(1,012), 10% validation (126), and 10% test (127). For the 32,413 NSCA patients, a 10-fold split is first applied due to computational limitations, and one fold containing 3,242 NSCA patients is randomly selected. These negative patients are then split into 80% training (2,593), 10% validation (324), and 10% test (325). After this, the NSCA patients in the training set are reduced to 1,012 so that the training set is balanced at the patient level. Undersampling is only applied to the training set, while the validation and test sets retain the natural class distribution. Because the split is based on patient ID, all visits from one patient remain in a single split, preventing cross-split leakage. For a fair comparison, only visits 1, 2, and 3 are considered for each patient, since most patients have only three visits. After this visit-level filtering, the classes are no longer exactly balanced because the number of eligible visits differs across patients and classes. The final training set contains 997 SCA and 1,568 NSCA visits, the validation set contains 125 SCA and 442 NSCA visits, and the test set contains 124 SCA and 419 NSCA visits.

B. Methods

This research aims to address the following question: How can we distinguish patients at risk of SCA from NSCA patients when positive SCA cases are limited? The main idea is to enhance LSTM-based prediction by incorporating a graph that links similar patient visits. Suppose that similar patient visits are related in meaningful ways that can be captured by a graph. This graph helps the LSTM become more robust, using information from similar visits and applying regularization to keep similar visits together while separating dissimilar ones. This structure is expected to improve prediction accuracy. The model also aims to identify NSCA patients accurately by minimizing false positives (FPs). The performance of the proposed method is evaluated by comparing various clustering and graph-construction strategies with a modified baseline LSTM based on the MIMIC-III benchmark model [18], using evaluation metrics, especially precision–recall area under the curve (PR-AUC). This metric is important due to the imbalance of SCA patients. Fig. 2 shows the model pipeline.

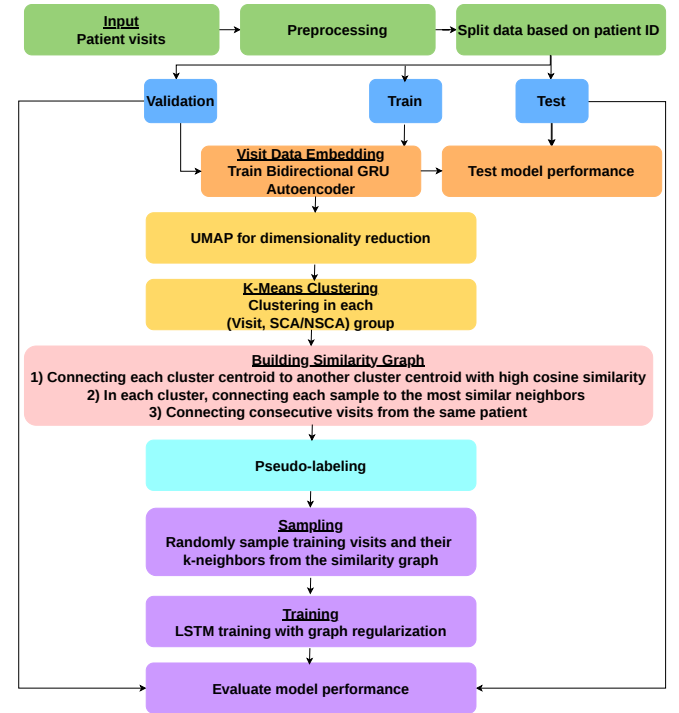


Fig. 2. SCA prediction pipeline. Visit-level time-series data are preprocessed and split by patient ID into training, validation, and test sets. A Bi-GRU autoencoder learns a compact embedding for each visit, and Uniform Manifold Approximation and Projection (UMAP) projects the embeddings into a lower-dimensional space. K-means is then applied separately by visit number and outcome group. The similarity graph connects related cluster centroids, similar visits within clusters, and consecutive visits from the same patient. The graph supports pseudo-labeling and neighbor selection for LSTM training with graph regularization, and final performance is evaluated on the test set.

C. Experimental Setup

a) AE Feature Embedding: The dataset includes visits with multivariate time-series data of varying lengths. An embedding method is used to convert them into fixed-length

20-dimensional representations while preserving temporal patterns. Traditional ML often summarizes longitudinal data, which can discard temporal dynamics for prediction [5]. In this work, a Bi-GRU AE is employed to learn a compact embedding of each complete sequence by capturing the long-term dependencies between time steps. This model encodes and decodes sequences using a bidirectional GRU, with the embedding vector z representing the entire sequence. The input consists of 76 features, and to standardize sequence lengths, all sequences are padded to the maximum length of 2198, which comes from the longest sequence in the training set after preprocessing. The Bi-GRU AE model thus processes sequences of equal length, with a Masking layer added after the Input layer to focus only on the actual data during training.

b) Clustering: After computing an embedding vector for each visit, the embeddings are clustered to identify meaningful groups. To improve clustering and reduce distances among same-class samples, the dataset is further stratified by visit ID and class label (SCA vs. NSCA). Then, K-means clustering with different values of K is applied within each group to minimize the within-cluster sum of squared distances. Clustering is performed using three representations: (1) the 20-dimensional latent vectors produced by the Bi-GRU AE; (2) a UMAP [20] embedding of the 20D latent vectors, with K-means applied in the UMAP space; and (3) a t-distributed stochastic neighbor embedding (t-SNE) [21] embedding of the 20D latent vectors, with K-means applied in the t-SNE space. For both UMAP and t-SNE, dimensionality reduction from 20 dimensions to 7, 10, and 15 is evaluated to identify the best clustering setting.

c) Graph Construction: After the clusters are identified, their centroids are used to build the graph. For each centroid, the cosine similarity with all the other centroids is calculated, and an edge is added when the similarity score is greater than 0.70. This threshold is chosen because it is the highest value that keeps the graph connected. Moreover, within each cluster, each visit is connected to its n most similar neighbors based on cosine similarity, where $n = |C|_{\min} - 1$ and $|C|_{\min}$ is the size of the smallest cluster. Finally, to capture how a patient's condition evolves over time, each visit is linked to the next visit of the same patient. This allows information to flow between similar visits and along the sequence of visits for each patient. The clustering results and similarity graph are then used to refine visit labels, generate pseudo-labels, and identify similar visits for training.

d) Pseudo-labeling: The constructed graph models the similarity between patient visits. One purpose of this graph is to identify borderline visits, meaning visits that are close to another group and may share some of its characteristics. In addition, the graph provides a second check on the ground-truth labels. For each visit, the weighted similarity scores of the positive and negative neighbors are calculated. Based on these scores, the visit is assigned a pseudo-label. These pseudo-labels are then used to train the final SCA model.

e) LSTM with Graph Regularization: In this paper, an LSTM model is used to capture the temporal relationship

within each visit and predict whether the visit is at risk of SCA. The patient-visit similarity graph guides data sampling for the LSTM model. The guided sampling method randomly selects a visit from the graph and augments it with its k -nearest neighbors based on cosine similarity. These visits are provided to the LSTM until all samples are included in training. The model is optimized using a hybrid loss function composed of two parts. The loss function is given in (1) [6].

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{Supervised}} + \lambda \mathcal{L}_{\text{Graph}}, \quad \lambda \in (0, 1). \quad (1)$$

$$\mathcal{L}_{\text{Supervised}} = \sum_{i=1}^B \mathcal{L}(y_i, \hat{y}_i) \quad (2)$$

$$\mathcal{L}_{\text{Graph}} = \sum_{i=1}^B \sum_{x_j \in \mathcal{N}(x_i)} w_{ij} \cdot D(h_{\theta}(x_i), h_{\theta}(x_j)) \quad (3)$$

The $\mathcal{L}_{\text{Supervised}}$ term in (2) is the supervised loss function, where B is the batch size, y_i represents the true label, $\hat{y}_i$ denotes the predicted output, and $\mathcal{L}$ is binary cross-entropy for binary classification. The $\mathcal{L}_{\text{Graph}}$ term in (3) is the unsupervised graph-embedding loss, where x_i is a sample, $\mathcal{N}(x_i)$ is the set of neighbors of x_i , and w_{ij} is the edge weight for the pair (x_i, x_j) , representing the cosine similarity calculated during graph construction. $h_{\theta}(x_i)$ and $h_{\theta}(x_j)$ are the embedding vectors of x_i and x_j learned by the model. D is the distance function that measures the cosine distance between the embeddings of neighboring samples. This term encourages similar visits to have similar representations by minimizing the cosine distance between neighboring visits. In (1), $\lambda \in (0, 1)$ controls the contribution of the graph regularization term relative to the supervised loss. A validation set is used to evaluate the model and tune hyperparameters during training. The trained model is applied to the test set.

IV. RESULTS AND DISCUSSION

A. AE Embedding Results

Fig. 3 illustrates the training loss curve, showing that the model can reconstruct the time-series data with no clear sign of overfitting. As the goal of this AE is to generate an embedding for each sample, reconstruction loss is used to evaluate model performance. Table I displays the mean squared error (MSE) for the train, validation, and test sets. It reports reconstruction performance and the final embedding shape.

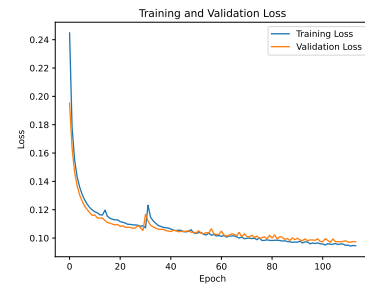


Fig. 3. Training and validation loss curves of Bi-GRU AE.

TABLE I
BI-GRU AE RECONSTRUCTION LOSS.

Dataset	MSE	Final embeddings shape
Train	0.094615	(2,565, 20)
Validation	0.097347	(567, 20)
Test	0.093970	(543, 20)

B. Clustering Results

Table II shows the silhouette scores for clustering in the original 20D latent space and 7D spaces produced by UMAP and t-SNE, where higher scores indicate better cluster cohesion and separation [22]. Based on dimensionality reduction from 20 to 7, 10, and 15 dimensions using UMAP and t-SNE, 7D gives the best results for both methods. Overall, clustering after dimensionality reduction performs better than clustering on the original embeddings. UMAP provides more consistent structure for the graph-construction setting, while t-SNE gives more variable results; therefore, t-SNE clustering is not used for graph construction.

TABLE II
SILHOUETTE SCORES FOR CLUSTERING ON ORIGINAL AND REDUCED EMBEDDINGS.

Approach	Visit	SCA	NSCA
K-means on original Bi-GRU AE embeddings (20D)	1	0.4220	0.2351
	2	0.3746	0.2332
	3	0.4448	0.2527
K-means on UMAP-reduced embeddings (7D)	1	0.7037	0.8695
	2	0.6168	0.8612
	3	0.4639	0.8819
K-means on t-SNE-reduced embeddings (7D)	1	0.4856	0.5848
	2	0.7015	0.2560
	3	0.8856	0.9027

C. LSTM Prediction Results

The experiments use an incremental comparison to show the effect of each added component. Because published SCA or CA studies use different cohorts, prediction windows, features, imbalance handling, and evaluation protocols, the standalone LSTM is used as a controlled baseline. Different model combinations are tested. (1) The baseline LSTM model predicts whether a patient visit is at risk of SCA. This model serves as the baseline, and the following models are built on it. (2) A graph-based model is used, in which K-means clustering is applied and a similarity graph is constructed. In this case, the minimum cluster size is 3 visits; therefore, the two most similar neighbors are passed to the LSTM with a graph regularization term for each sample. The LSTM learns similar embeddings for neighboring samples. (3) The approach is similar to method (2), but UMAP is applied before clustering to reduce dimensionality and improve clustering quality. The minimum cluster size is 10, allowing up to 9 neighbors to be sampled and passed to the LSTM. Table III compares these models under different neighborhood sizes ($Neighbors \in \{2, 5, 7, 9\}$). For each model, several λ values are tested, and the value with the highest validation PR-AUC is reported. Temperature scaling is applied to the similarity weights because many cosine similarity scores are close to each other. By sharpening or flattening the softmax distribution, it controls how strongly the graph loss emphasizes the most similar neighbors. To address class imbalance, class weighting is applied during training.

Because false negatives (FNs) may have important clinical consequences, PR-AUC is used as the primary model selection metric. Based on PR-AUC, the UMAP-based similarity-graph model with 9 neighbors and temperature scaling (Model 11) performs best with PR-AUC = 0.5968, outperforming both the baseline LSTM (PR-AUC = 0.5731) and the similarity-graph models without UMAP. Compared with the baseline model (Model 1), Model 11 performs better on all metrics except recall (0.6129 vs. 0.8065). Moreover, Model 11 produces better results than the graph-based models without UMAP, suggesting that UMAP improves the similarity structure used for clustering and graph regularization. The confusion matrices on the test set for the baseline model (Model 1) and UMAP-based model with scaling (Model 11) are shown in Fig. 4. The UMAP-based model with scaling identifies NSCA cases more effectively, resulting in fewer FPs and higher overall accuracy than the baseline model. However, Model 11 reduces recall, meaning that more SCA visits are missed than with the baseline. Since all models are selected using validation PR-AUC and evaluated at a fixed threshold of 0.5, recall reflects one operating point rather than the full ranking behavior of the model. The lower recall suggests that fewer SCA visits receive probabilities above 0.5. This may be related to the strong graph regularization ($\lambda=0.9$), which encourages similar visits to have closer representations. Because the graph is based on embedding similarity rather than outcome labels only, some borderline SCA visits may be regularized toward neighboring NSCA-like visits, reducing FPs but increasing FNs.

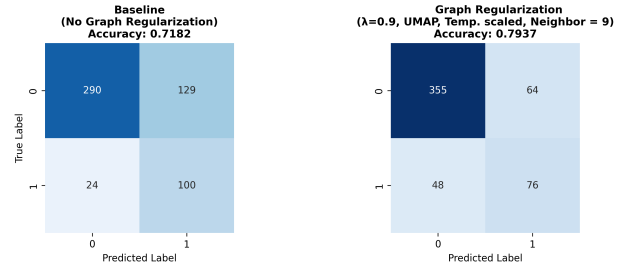


Fig. 4. Test-set confusion matrices for the baseline and graph-based SCA prediction models. The graph-based model ($\lambda = 0.9$, UMAP with temperature scaling, 9 neighbors) reduces FPs while recall decreases.

V. CONCLUSION, LIMITATIONS, AND FUTURE WORK

An LSTM model combined with graph regularization was developed to identify patient visits at risk of SCA. For each visit, the multivariate time-series was embedded using a Bi-GRU AE, then reduced in dimension and clustered with K-means. A cosine-similarity graph was constructed to connect similar visits, and the k -nearest neighbors from this graph were used to regularize LSTM training through a graph-based loss term in addition to binary cross-entropy. By combining temporal modeling with similarity information among visits, the proposed approach improved prediction performance on this highly imbalanced task. In particular, the model using UMAP for dimensionality reduction outperformed the baseline LSTM, achieving gains of 10.5% in accuracy, 24.3% in

TABLE III

TEST-SET PERFORMANCE OF THE BASELINE AND SIMILARITY-GRAPH MODELS SELECTED BY VALIDATION PR-AUC USING $\mathcal{L} = \mathcal{L}_{\text{Supervised}} + \lambda \mathcal{L}_{\text{Graph}}$. COMPARED WITH MODEL 1, MODEL 11 IMPROVES ACCURACY BY 10.5%, PRECISION BY 24.3%, F1-SCORE BY 1.6%, ROC-AUC BY 1.9%, AND PR-AUC BY 4.1%, WHILE RECALL DECREASES BY 24.0%.

Number	Model	Neighbors	λ_{weight}	Temperature scaling	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
1	LSTM	—	—	—	0.7182	0.4367	0.8065	0.5666	0.8091	0.5731
2	LSTM + K-means + Similarity Graph	2	0.5	No	0.7643	0.4853	0.5323	0.5077	0.7913	0.5605
3	LSTM + K-means + Similarity Graph	2	0.7	Yes	0.7495	0.4655	0.6532	0.5436	0.8177	0.5832
4	LSTM + UMAP + K-means + Similarity Graph	2	0.9	No	0.7422	0.4583	0.7097	0.5570	0.8075	0.5741
5	LSTM + UMAP + K-means + Similarity Graph	2	0.2	Yes	0.7956	0.5496	0.5806	0.5647	0.7699	0.5817
6	LSTM + UMAP + K-means + Similarity Graph	5	0.5	No	0.7440	0.4595	0.6855	0.5502	0.7657	0.5294
7	LSTM + UMAP + K-means + Similarity Graph	5	0.1	Yes	0.7182	0.4293	0.7097	0.5350	0.7904	0.5509
8	LSTM + UMAP + K-means + Similarity Graph	7	0.5	No	0.7385	0.4511	0.6694	0.5390	0.7914	0.5603
9	LSTM + UMAP + K-means + Similarity Graph	7	0.3	Yes	0.7440	0.4561	0.6290	0.5288	0.8114	0.5887
10	LSTM + UMAP + K-means + Similarity Graph	9	0.7	No	0.7477	0.4641	0.6774	0.5508	0.8078	0.5868
11	LSTM + UMAP + K-means + Similarity Graph	9	0.9	Yes	0.7937	0.5429	0.6129	0.5758	0.8245	0.5968

precision, 1.6% in F1-score, and 4.1% in PR-AUC. These results suggest that incorporating inter-visit similarity structure can enhance risk prediction beyond sequence modeling alone, and that UMAP is effective for preserving this structure in the learned representation space. However, this study has limitations. First, only the first three visits were considered because most patients had no more than three visits. Second, SCA visits were relatively rare, accounting for only 3.05% of all visits, resulting in a highly imbalanced dataset. Simple oversampling of SCA visits was not used because the graph representation assigns one node to each visit, and duplicating visits could distort the similarity graph. Future work should examine time-series augmentation methods to increase minority-class information without creating duplicate nodes. Third, due to computational constraints, only one of the 10 NSCA folds was used in the splitting strategy. Future studies should evaluate the remaining folds, include all available data, and report average results across folds. In addition, alternative clustering methods could be explored. Transfer learning with pre-trained models could also be tested as an alternative to the LSTM baseline. Moreover, future work could examine lower decision thresholds, apply recall-constrained model selection, or incorporate outcome labels into graph construction to recover sensitivity.

REFERENCES

- [1] S. S. Martin, A. W. Aday, N. B. Allen, Z. I. Almarzooq, C. A. Anderson, P. Arora, C. L. Avery, C. M. Baker-Smith, N. Bansal, A. Z. Beaton *et al.*, “2025 heart disease and stroke statistics: a report of US and global data from the American Heart Association,” *Circulation*, vol. 151, no. 8, pp. e41–e660, 2025.
- [2] O. Almousa, J. Prates, N. Yeslam, D. Mac Gregor, J. Zhang, V. Phan, M. Nielsen, R. Smith, and K. Qayumi, “Virtual reality simulation technology for cardiopulmonary resuscitation training: An innovative hybrid system with haptic feedback,” *Simulation & Gaming*, vol. 50, no. 1, pp. 6–22, 2019.
- [3] S. Aqel, S. Syaj, A. Al-Bzour, F. Abuzanounh, N. Al-Bzour, and J. Ahmad, “Artificial intelligence and machine learning applications in sudden cardiac arrest prediction and management: a comprehensive review,” *Current Cardiology Reports*, vol. 25, no. 11, pp. 1391–1396, 2023.
- [4] K. Häyrynen, K. Saranto, and P. Nykänen, “Definition, structure, content, use and impacts of electronic health records: a review of the research literature,” *International Journal of Medical Informatics*, vol. 77, no. 5, pp. 291–304, 2008.
- [5] F. Xie, H. Yuan, Y. Ning, M. E. H. Ong, M. Feng, W. Hsu, B. Chakraborty, and N. Liu, “Deep learning for temporal data representation in electronic health records: A systematic review of challenges and methodologies,” *Journal of Biomedical Informatics*, vol. 126, p. 103980, 2022.
- [6] T. D. Bui, S. Ravi, and V. Ramavajjala, “Neural graph learning: Training neural networks using graphs,” in *Proceedings of the Eleventh ACM International Conference on Web Search and Data Mining*, 2018, pp. 64–71.
- [7] S. L. Javan, M. M. Sepehri, M. L. Javan, and T. Khatibi, “An intelligent warning model for early prediction of cardiac arrest in sepsis patients,” *Computer Methods and Programs in Biomedicine*, vol. 178, pp. 47–58, 2019.
- [8] B. Soudan, F. F. Dandachi, and A. B. Nassif, “Attempting cardiac arrest prediction using artificial intelligence on vital signs from electronic health records,” *Smart Health*, vol. 25, p. 100294, 2022.
- [9] L. Yijing, Y. Wenyu, Y. Kang, Z. Shengyu, H. Xianliang, J. Xingliang, W. Cheng, S. Zehui, and L. Mengxing, “Prediction of cardiac arrest in critically ill patients based on bedside vital signs monitoring,” *Computer Methods and Programs in Biomedicine*, vol. 214, p. 106568, 2022.
- [10] I. Gandin, A. Scagnetto, S. Romani, and G. Barbati, “Interpretability of time-series deep learning models: A study in cardiovascular patients admitted to intensive care unit,” *Journal of Biomedical Informatics*, vol. 121, p. 103876, 2021.
- [11] M. Ma, X. Hao, J. Zhao, S. Luo, Y. Liu, and D. Li, “Predicting heart failure in-hospital mortality by integrating longitudinal and category data in electronic health records,” *Medical & Biological Engineering & Computing*, vol. 61, no. 7, pp. 1857–1873, 2023.
- [12] H. O. Boll, A. Amirahmadi, A. Soliman, S. Byttner, and M. Recamonde-Mendoza, “Graph neural networks for heart failure prediction on an EHR-based patient similarity graph,” *arXiv preprint arXiv:2411.19742*, 2024.
- [13] I. Alrashdi and A. I. Taloba, “Integration of graph neural networks and long short-term memory models for advancing heart failure prediction,” *Alexandria Engineering Journal*, vol. 127, pp. 143–163, 2025.
- [14] L. Lyu, H. Zhu, H. Chen, J. Zhou, R. H. Chan, and L. Lu, “PDSNet: Patient-disease dual spatial similarity neural networks for predicting heart failure risk using short electronic health records,” *IEEE Journal of Biomedical and Health Informatics*, 2025.
- [15] K. Berahmand, F. Daneshfar, E. S. Salehi, Y. Li, and Y. Xu, “Autoencoders and their applications in machine learning: a survey,” *Artificial Intelligence Review*, vol. 57, no. 2, p. 28, 2024.
- [16] A. Likas, N. Vlassis, and J. J. Verbeek, “The global k-means clustering algorithm,” *Pattern Recognition*, vol. 36, no. 2, pp. 451–461, 2003.
- [17] A. E. Johnson, T. J. Pollard, L. Shen, L.-w. H. Lehman, M. Feng, M. Ghassemi, B. Moody, P. Szolovits, L. A. Celi, and R. G. Mark, “MIMIC-III, a freely accessible critical care database,” *Scientific Data*, vol. 3, no. 1, pp. 1–9, 2016.
- [18] H. Harutyunyan, H. Khachatrian, D. C. Kale, G. Ver Steeg, and A. Galstyan, “Multitask learning and benchmarking with clinical time series data,” *Scientific Data*, vol. 6, no. 1, p. 96, 2019.
- [19] R. Afrin, E. A. Mohammed, and B. Far, “Visual representation of tabular electronic health records for predicting sudden cardiac arrest,” in *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE, 2024, pp. 1–7.
- [20] L. McInnes, J. Healy, and J. Melville, “UMAP: Uniform manifold approximation and projection for dimension reduction,” *arXiv preprint arXiv:1802.03426*, 2018.
- [21] L. Van der Maaten and G. Hinton, “Visualizing data using t-SNE,” *Journal of Machine Learning Research*, vol. 9, no. 11, 2008.
- [22] K. R. Shahapure and C. Nicholas, “Cluster quality analysis using silhouette score,” in *2020 IEEE 7th International Conference on Data Science and Advanced Analytics (DSAA)*. IEEE, 2020, pp. 747–748.